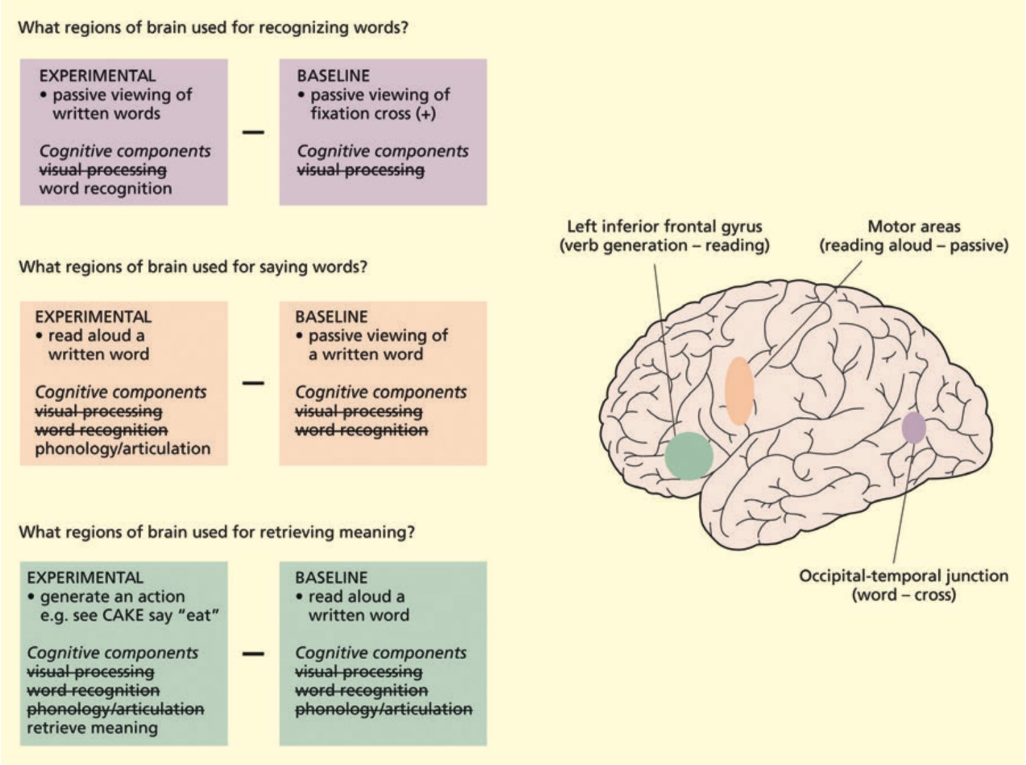
Machine learning methods in psychology & neuroscience

Professor Maureen Ritchey

How do we usually analyze brain activity data?

The cognitive subtraction method for analyzing fMRI data



How do we usually analyze brain activity data?

To do these analyses, we typically need to know what the participant is thinking or experiencing at a given moment in time.

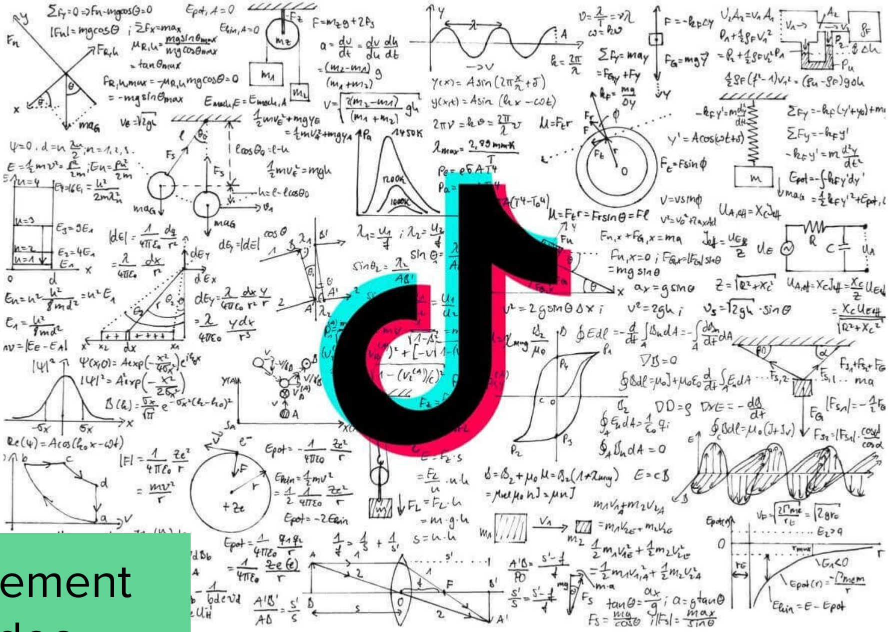
What if we don’t?

What if we want to guess?

What is machine learning?

What is machine learning?

- An algorithm that learns to make predictions: given a new set of data, similar to other data it has seen before, what will be the outcome?
- It does this by identifying patterns in the data and then making inferences based on those patterns



d.io/learn/reverse-engineering-how-tiktok-algorithm-works

Dataset: Video features, other users' engagement
Outcomes: Whether you engage with the video
Predictions: Will you like this new video?

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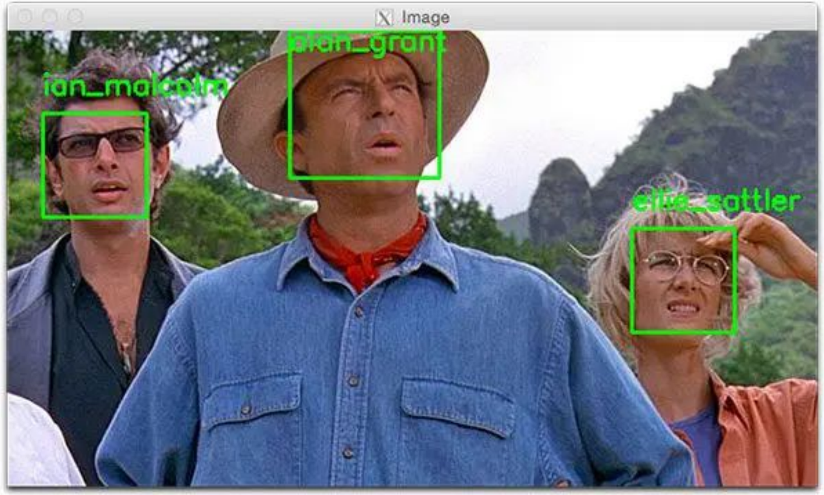


https://pyimagesearch.com/2018/06/18/face-recognition-with-opencv-python-and-deep-learning/

Dataset: Image features
Outcomes: Whether the image contains a face
Predictions: Does this new image contain a face?

How do we use methods like these to study the brain?

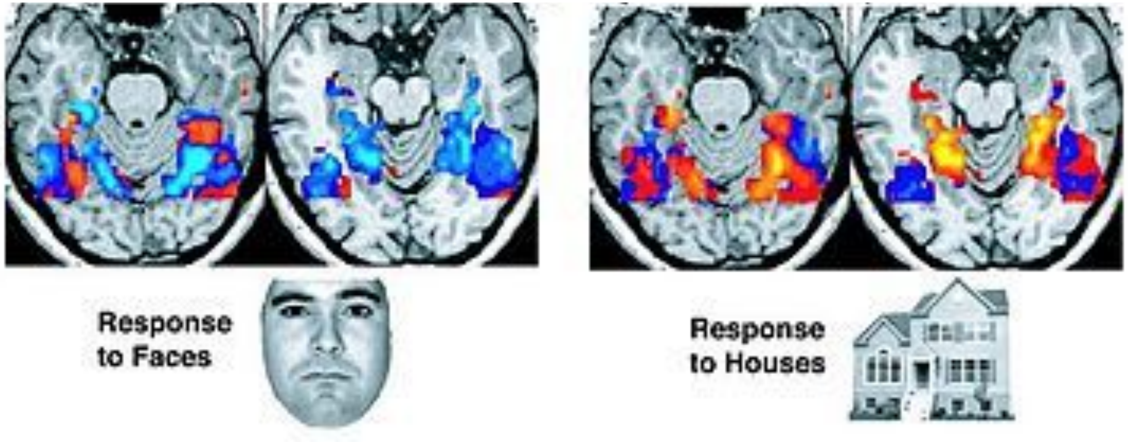
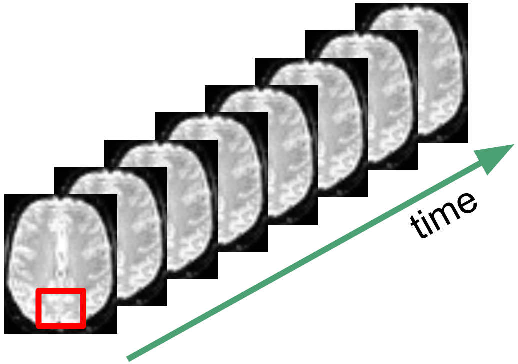
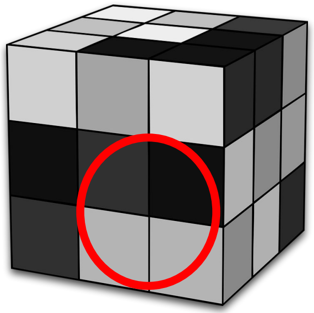


Figure from Haxby et al. 2001

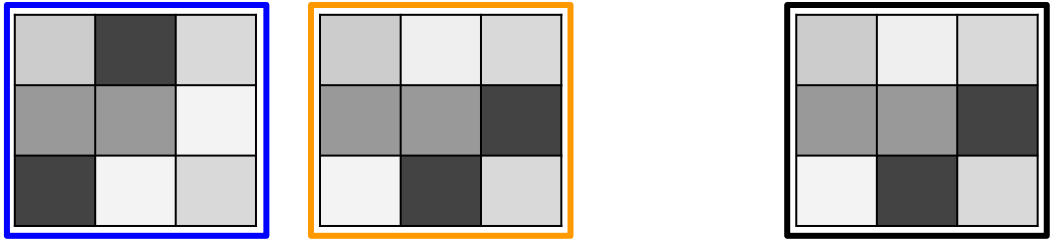
Dataset: Brain activity patterns
Outcomes: What the participant was seeing or thinking about
Predictions: What are they seeing or thinking about now?

Multi-voxel pattern analysis
An application of machine learning methods in fMRI





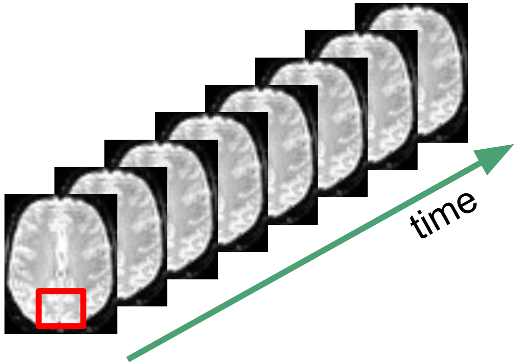
We can look at how the spatial **pattern** of the BOLD signal is affected by our experimental conditions —> multi-voxel pattern analysis



Condition A   Condition B   Which?

Average activity is the same!
Only the pattern differs.

Multi-voxel pattern analysis
An application of machine learning methods in fMRI



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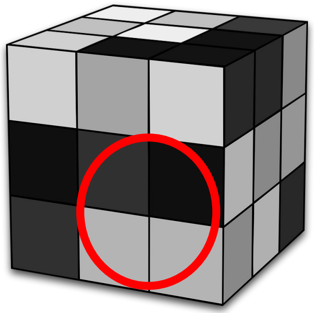
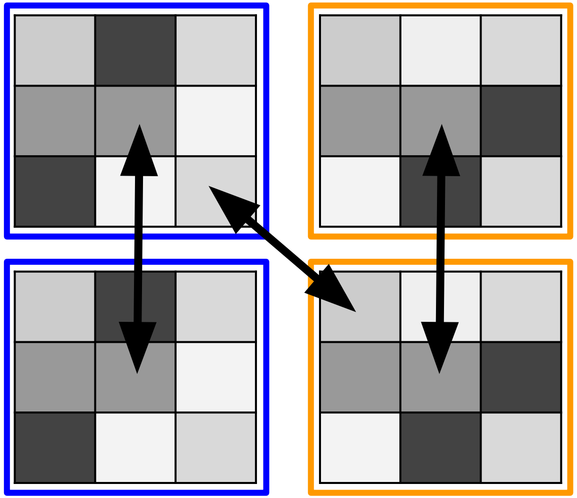


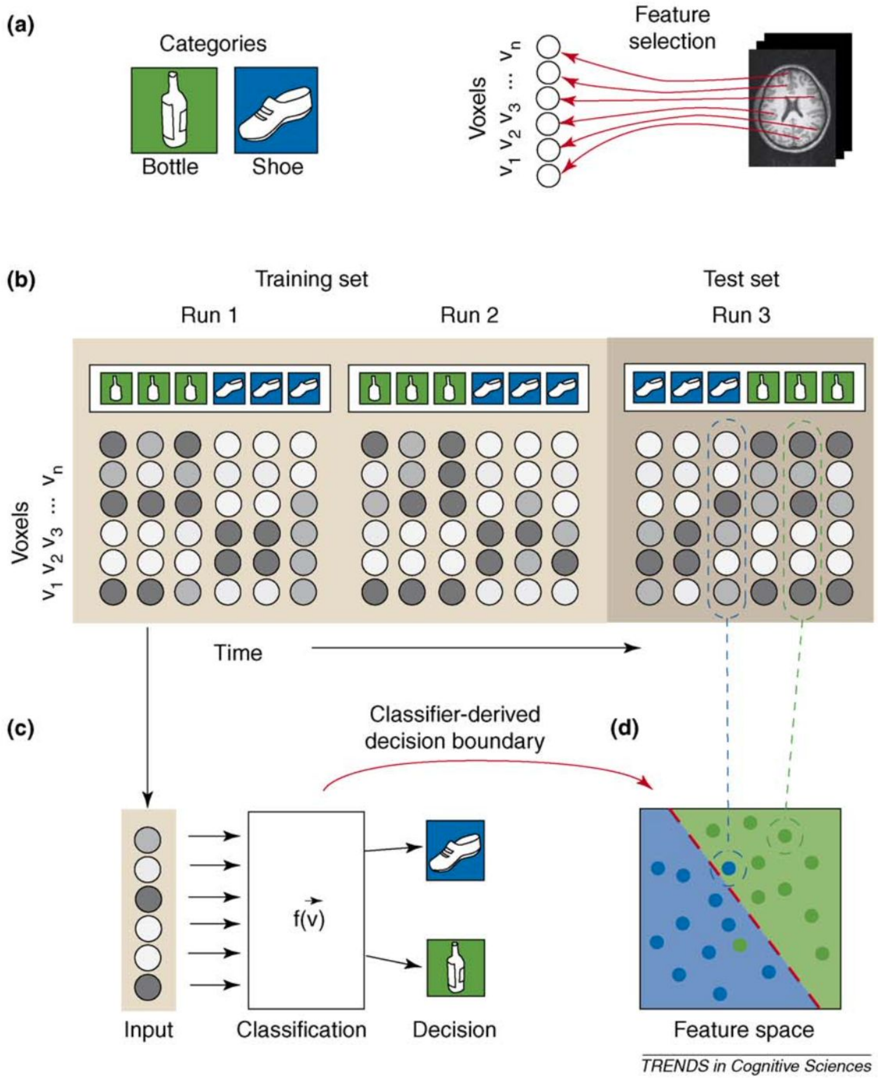
corr(A,A) > corr(A,B)

corr(B,B) > corr(A,B)



What are some potential applications of these kinds of methods?

Getting into the details



Important terms:
Feature selection: which data do you want to use to make predictions?
Classifier: the machine learning algorithm that links features of the dataset to their outcomes (labels)
Training & testing sets: which data do you use to teach the classifier the relationship between features and labels, and which do you use to test if you were right?
Decision boundary: the "line" that separates the data that lead to one label from those that lead to the other label
*technically a boundary in *n*-dimensional space, where *n*=number of features
Support vector machines, artificial neural networks, logistic regression: different kinds of classifier algorithms

Using MVPA to track memory recall

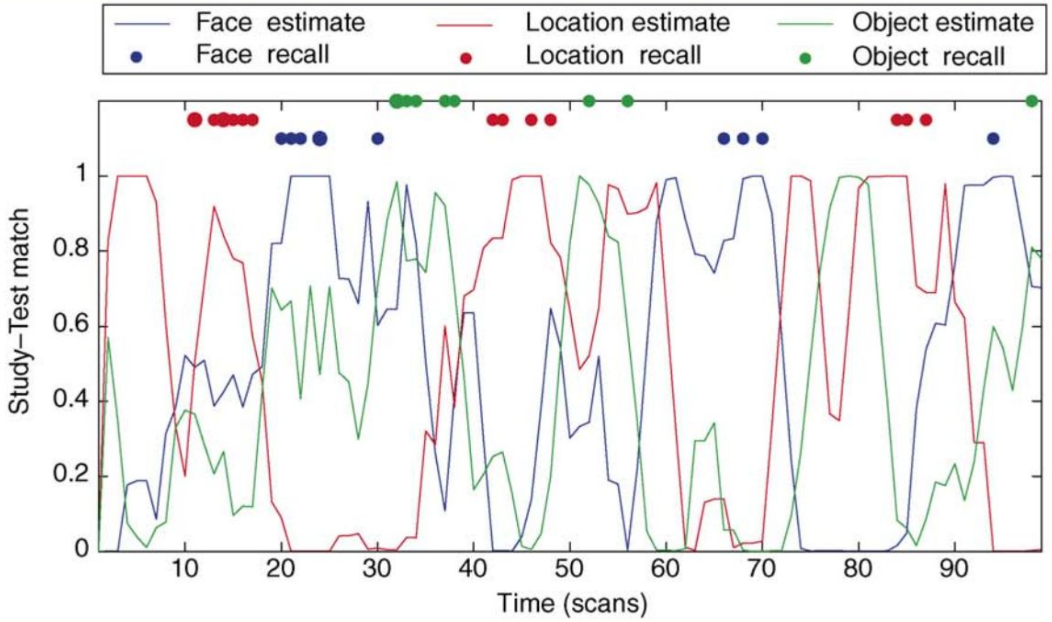


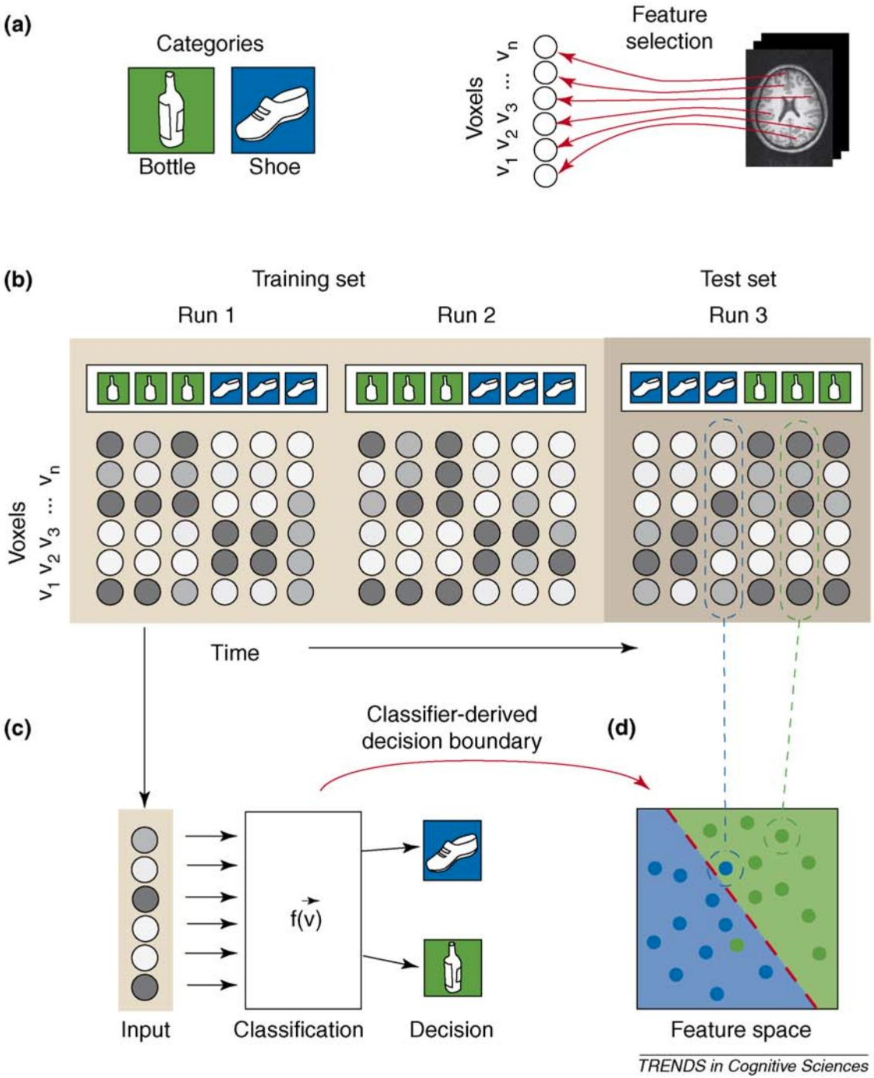
Figure I. Illustration of how brain activity during recall relates to recall behavior, in a single subject. Each point on the *x*-axis corresponds to a 1.8 s interval (during the 3-min recall period). The blue, red, and green lines correspond to the classifier's estimate as to how strongly the subject is reinstating brain patterns characteristic of face-study, location-study, and object-study at that point in time. The blue, red, and green dots indicate time points where subjects recalled faces, locations, and objects; the dots were shifted forward by three time-points, to account for the lag in the peak hemodynamic response. The graph illustrates the strong correspondence between the classifier's estimate of category-specific brain activity, and the subject's actual recall behavior. (Reprinted with permission from [29].)

But wait, there's more

Machine learning methods are not restricted to fMRI - you can use the same type of approach to decode brain activity measured with EEG, intracranial recordings, or any other measure that gives you lots of observations.

You can also use it to build models of people's **behavior**, for example:

- Can we predict whether or not someone will remember a particular image? If so, what info do we need to make this prediction?
- Can we predict whether or not someone has depression based on their texting behavior?
- Can we predict who will be compatible as romantic partners based on a set of personality questionnaires?



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Support vector machines, artificial neural networks, logistic regression: different kinds of classifier algorithms

Is Romantic Desire Predictable? Machine Learning Applied to Initial Romantic Attraction

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Abstract

Matchmaking companies and theoretical perspectives on close relationships suggest that initial attraction is, to some extent, a product of two people's self-reported traits and preferences. We used machine learning to test how well such measures predict people's overall tendencies to romantically desire other people (actor variance) and to be desired by other people (partner variance), as well as people's desire for specific partners above and beyond actor and partner variance (relationship variance). In two speed-dating studies, romantically unattached individuals completed more than 100 self-report measures about traits and preferences that past researchers have identified as being relevant to mate selection. Each participant met each opposite-sex participant attending a speed-dating event for a 4-min speed date. Random forests models predicted 4% to 18% of actor variance and 7% to 27% of partner variance; crucially, however, they were unable to predict relationship variance using any combination of traits and preferences reported before the dates. These results suggest that compatibility elements of human mating are challenging to predict before two people meet.